Stop and wait protocol-Part IV

Comprehensive Course on Computer Networks

Ankit Kumar • Lesson 38 • Apr 14, 2026

unacademy

CS & IT Engineering

Computer Network

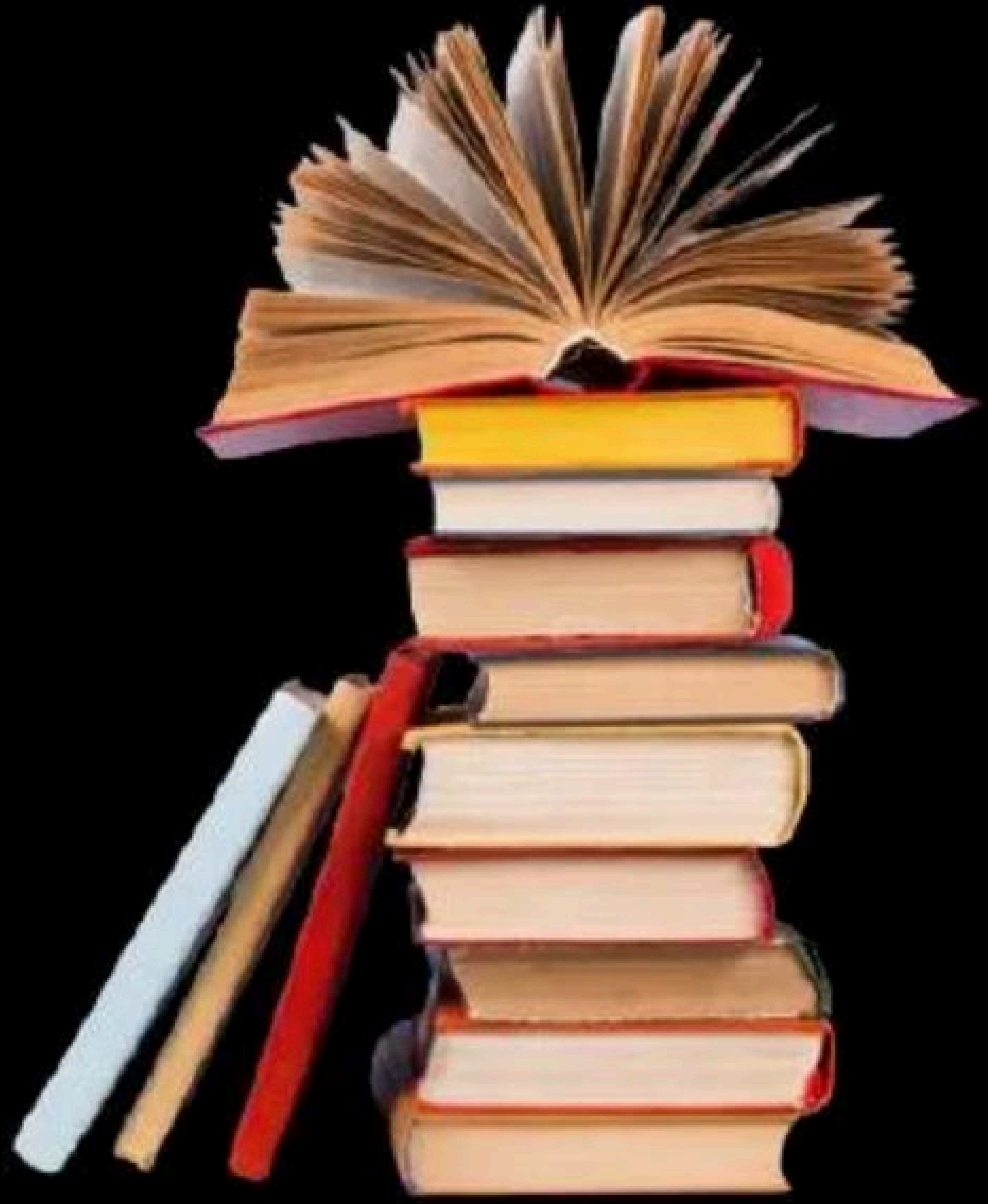
Error Control

Lecture Number

By- Ankit Sir

Topics

to be covered

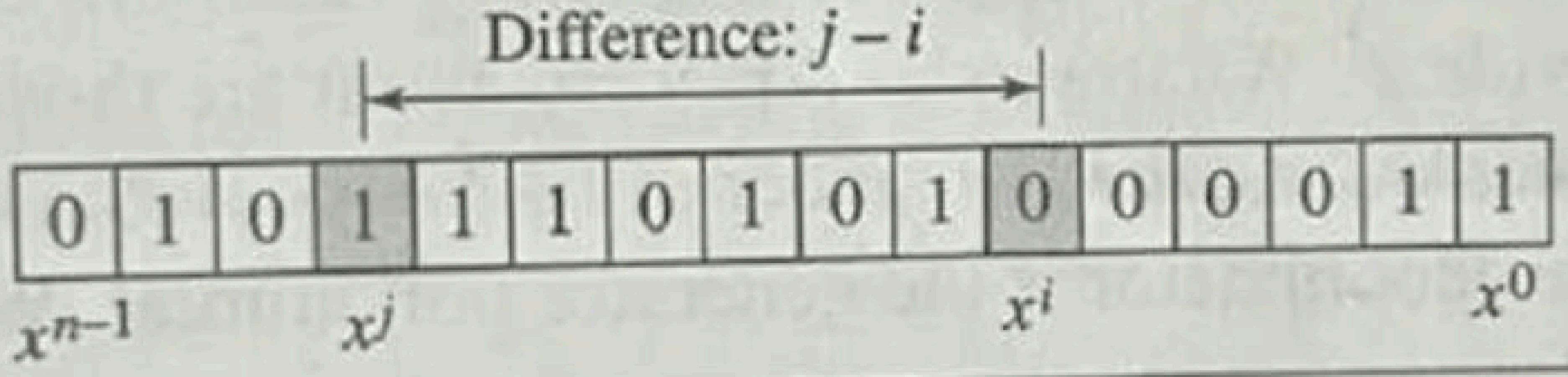


CRC

Two Isolated Single-Bit Errors

Now imagine there are two single-bit isolated errors. Under what conditions can this type of error be caught? We can show this type of error as $e(x) = x^j + x^i$. The values of i and j define the positions of the errors, and the difference $j - i$ defines the distance between the two errors, as shown in Figure 10.10.

Figure 10.10 *Representation of two isolated single-bit errors using polynomials*



$E(x) = x^i + x^j$

$= \quad x5$

We can write $e(x) = x^i(x^{j-i} + 1)$. If $g(x)$ has more than one term and one term is x^0, it cannot divide x^i, as we saw in the previous section. So if $g(x)$ is to divide $e(x)$, it must divide $x^{j-i} + 1$. In other words, $g(x)$ must not divide $x^t + 1$, where t is between 0 and $n - 1$. However, $t = 0$ is meaningless and $t = 1$ is needed, as we will see later. This means t should be between 2 and $n - 1$.

If a generator cannot divide $x^t + 1$ (t between 0 and $n - 1$), then all isolated double errors can be detected.

Find the status of the following generators related to two isolated, single-bit errors.

a) $x + 1$

b) $x^4 + 1$

c) $x^7 + x^6 + 1$

d) $x^{15} + x^{14} + 1$

Next

Solution

a. This is a very poor choice for a generator. Any two errors next to each other cannot be detected.

b. This generator cannot detect two errors that are four positions apart. The two errors can be anywhere, but if their distance is 4, they remain undetected.

c. This is a good choice for this purpose.

Stop and wait protocol-Part IV

Comprehensive Course on Computer Networks

Ankit Kumar • Lesson 38 • Apr 14, 2026

unacademy

CS & IT Engineering

Computer Network

Error Control

Lecture Number

By- Ankit Sir

- ❑ All burst errors with $L \leq r$ will be detected.
- ❑ All burst errors with $L = r + 1$ will be detected with probability $1 - (1/2)^{r-1}$.
- ❑ All burst errors with $L > r + 1$ will be detected with probability $1 - (1/2)^{r}$.

Example 10.10

Find the suitability of the following generators in relation to burst errors of different lengths.

~~a.~~ $x^6 + 1$

b. $x^{18} + x^7 + x + 1$

c. $x^{32} + x^{23} + x^7 + 1$

Q10-11. In CRC, we have chosen the generator 1100101. What is the probability of detecting a burst error of length

a. 5?
b. 7?
c. 10?

Q10-12. In CRC, which of the following generators

Consider the CRC polynomial $x^8 + x^7 + x^6 + x^4 + x^2 + 1$. Which of the following is false ?

A. It detects a single bit error

B. it detects a burst error of size 6

C. The probability of detecting a burst error of size 9 is $\frac{1}{2^7}$

D. The probability of detecting a burst error of size 15 is $\frac{1}{2^{15}}$

Standard Polynomials

Some standard polynomials used by popular protocols for CRC generation are shown in Table 10.4 along with the corresponding bit pattern.

Table 10.4 *Standard polynomials*

Name	*Polynomial*	*Used in*
CRC-8	$x^8 + x^2 + x + 1$ 100000111	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$ 11000110101	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$ 10001000000100001	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$ 100000100110000010001110110110111	LANs

1. If the generator has more than one term and coefficient of x^0 is 1, all single bit error can be detected.
2. If a generator cannot divide $x^t + 1$ (t between 0 and $n - 1$) then all isolated Double error can be detected
3. A generator that contains a Factor of $x + 1$ can detect all odd numbered errors.

A good polynomial generator needs to have the following characteristics:

1. It should have at least two terms.
2. The coefficient of the term x^0 should be 1.
3. It should not divide $x^t + 1$, for t between 2 and n - 1.
4. It should have the factor $x + 1$.

Topic: Today's Summary

THANK YOU!

Here's to a cracking journey ahead!